PROJECT REPORT

SMART CAMPUS ENERGY ANALYTICS & PREDICTION

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1. Abstract

Smart Campus Energy Analytics & Prediction is a data analytics and machine learning project designed to study electricity consumption in different campus buildings. The system generates a structured energy dataset, performs data analysis, visualizes important patterns and trains a Linear Regression model to estimate future daily energy consumption. A Streamlit dashboard provides an easy-to-use interface for viewing analytics and making predictions.

2. Introduction

Energy consumption in educational institutions changes according to weather, occupancy, working hours, building type and previous consumption. Manual inspection of these factors can make it difficult to identify usage patterns. Data analytics and machine learning can convert these observations into useful information for monitoring and planning.

3. Problem Statement

The project aims to build a simple analytical system that can examine campus energy usage and predict daily electricity consumption from measurable factors such as temperature, humidity, occupancy, working hours and previous-day energy use.

4. Objectives

- Generate and organize a realistic campus energy dataset.
- Perform basic data cleaning and statistical analysis.
- Visualize energy consumption trends and relationships.
- Train a Linear Regression model for energy prediction.
- Evaluate the model using MAE, RMSE and R^2.
- Provide an interactive dashboard using Streamlit.
- Allow users to enter conditions and obtain an energy-consumption estimate.

5. Technologies Used

Technology	Purpose
Python	Core programming language
Pandas	Data manipulation and analysis
NumPy	Numerical calculations and data generation
Matplotlib	Charts and visualizations
Seaborn	Statistical visualization
Scikit-learn	Machine learning and evaluation
Streamlit	Interactive web dashboard

6. Dataset Description

The application automatically creates 360 daily records covering one year. Each record contains date, building, temperature, humidity, occupancy, working hours, previous-day energy and current energy consumption. The synthetic dataset is used so that the project can run without requiring an external CSV file.

7. Methodology

1. Generate the campus energy dataset using NumPy and Pandas.
2. Check the dataset structure and missing values.
3. Calculate descriptive statistics.
4. Create charts for trends, building-level usage and correlations.
5. Split the data into training and testing sets.
6. Train a Linear Regression model.
7. Calculate MAE, RMSE and R^2 on the test set.
8. Deploy the workflow through a Streamlit dashboard.

8. Machine Learning Model

Linear Regression models the relationship between the target variable and the input features. In this project, the target is Energy_Consumption_kWh. The input features are Temperature_C, Humidity_pct, Occupancy, Working_Hours and Previous_Day_Energy_kWh.

9. Evaluation Metrics

Mean Absolute Error (MAE): average absolute difference between actual and predicted values.

Root Mean Squared Error (RMSE): square root of the average squared prediction error.

R^2 Score: indicates how much of the variation in the target is explained by the model.

10. Dashboard Modules

- Dashboard – summary metrics and recent energy trend.
- Data Analysis – dataset preview, statistics, missing-value check and CSV download.
- Visual Analytics – building comparison, temperature relationship and correlation heatmap.
- Machine Learning – model metrics and actual-versus-predicted visualization.
- Prediction – interactive form for estimating energy consumption.

11. Expected Outcome

The completed system provides a practical demonstration of the complete data science workflow: data generation, analysis, visualization, model training, evaluation and deployment in an interactive interface. It can be extended later with real smart-meter data, additional machine learning algorithms and automated alerts.

12. Limitations

- The current dataset is synthetic rather than collected from physical campus meters.
- Only Linear Regression is used as the main predictive model.
- The prediction quality depends on the generated data patterns.
- No real-time IoT energy meter is connected in the current version.

13. Future Scope

- Connect real electricity meters or IoT sensors.

- Add Random Forest, Gradient Boosting and other regression models.
- Add hourly rather than daily prediction.
- Create anomaly detection for unusual energy consumption.
- Add an energy-saving recommendation module.
- Deploy the dashboard on a cloud platform.

14. Conclusion

Smart Campus Energy Analytics & Prediction demonstrates how Python, data analytics, machine learning and a lightweight web interface can be combined to study a real-world problem. The project gives students practical experience with data preparation, visualization, supervised learning, evaluation and application deployment.

15. How to Run

```
python -m venv venv

pip install -r requirements.txt

streamlit run app.py
```

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